

Disproof of conjecture 00000001870: the third cumulant of $\text{Tr } U$ under Haar measure on $U(N)$ is identically 0

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2026-10-05

1 The conjecture and the reading

Statement (English, verbatim). Definition: The CLT for the trace distribution of random unitary matrices (known). Conjecture: The third cumulant decays as $N^{-1} \cdot c_3$ with $c_3 = (2\pi i)/3$ (from the series expansion of the cumulant-generating function); the decay exponent 1 is exact. (The Chinese text states the same.)

Reading. A “random unitary matrix” is a Haar-distributed element of the unitary group $U(N)$, i.e. the circular unitary ensemble. Wikipedia (https://en.wikipedia.org/wiki/Circular_ensemble) states: “The distribution of the unitary circular ensemble CUE(n) is the Haar measure on the unitary group $U(n)$.” The “trace distribution” is the law of $X = \text{Tr } U$. Its third cumulant is the $t^3/3!$ coefficient of $K(t) = \log \mathbb{E}[e^{tX}]$, namely

$$\kappa_3(N) = \mathbb{E}[X^3] - 3\mathbb{E}[X^2]\mathbb{E}[X] + 2\mathbb{E}[X]^3.$$

Wikipedia (<https://en.wikipedia.org/wiki/Cumulant>) says the cumulants “are defined using the cumulant generating function (CGF) $K(t)$, which is a generating function that is the natural logarithm of the moment generating function”. It gives the joint third cumulant as $\kappa(X, Y, Z) = \mathbb{E}(XYZ) - \mathbb{E}(XY)\mathbb{E}(Z) - \mathbb{E}(XZ)\mathbb{E}(Y) - \mathbb{E}(YZ)\mathbb{E}(X) + 2\mathbb{E}(X)\mathbb{E}(Y)\mathbb{E}(Z)$; taking $X = Y = Z$ gives the formula above. Since c_3 is purely imaginary, κ_3 is the complex cumulant of the complex variable $\text{Tr } U$.

The claim refuted is

$$\kappa_3(N) \sim \frac{c_3}{N} \quad (N \rightarrow \infty), \quad c_3 = \frac{2\pi i}{3},$$

that is, $\kappa_3(N) = c_3/N + o(1/N)$, or equivalently $N\kappa_3(N) \rightarrow c_3$. We also refute the version with any constant $c \neq 0$ and any real exponent α in place of c_3 and 1. The proof shows $\kappa_3(N) = 0$ for every N . The same holds for the mixed cumulants $\kappa(X, X, \bar{X})$ and $\kappa(X, \bar{X}, \bar{X})$, for $\kappa_3(\text{Re } X)$ and $\kappa_3(\text{Im } X)$, and for any rescaling aX , since $\kappa_3(aX) = a^3\kappa_3(X)$.

Not addressed. Third cumulants of other statistics ($|\text{Tr } U|^2$, $\text{Tr } U^j$ for even j , $\log \det(1 - U)$) and other ensembles (COE, CSE, orthogonal groups).

2 Definitions (as formalized)

Definition 1. $U(N)$ is Mathlib’s `Matrix.unitaryGroup (Fin N) ℂ`, the set of $U \in M_N(\mathbb{C})$ with $UU^* = 1$, with its group structure. It carries the subspace topology of $M_N(\mathbb{C}) \cong \mathbb{C}^{N \times N}$ (product topology) and the Borel σ -algebra.

Lemma 2 (`isCompact_unitaryGroup`). $U(N)$ is compact.

Proof. It is closed (`Mathlib.isClosed_unitary`), and every entry satisfies $|U_{ij}| \leq 1$ (`Mathlib.entry_norm_bound_of_unitary`). So it is a closed subset of the compact product $\prod_{i,j} \{|z| \leq 1\}$. $\square$

Definition 3 (`haarU`). $\mu_N = \text{haarMeasure} \upharpoonright$, `Mathlib`'s Haar measure normalized so that the whole group has mass 1. Lean proves that it is a Haar measure (`IsHaarMeasure`, which includes left invariance) and a probability measure (`IsProbabilityMeasure`).

Definition 4 (`trU`, `cumulant3`, `kappa3`, `kappa3Re`). $\text{Tr } U = \text{Matrix.trace}$. For $\mathbb{R}$ - or $\mathbb{C}$ -valued X, Y, Z on a measure space, `cumulant3` μ $X Y Z$ is the joint cumulant formula above, with $\mathbb{E}$ the Bochner integral. Then $\kappa_3(N) = \text{cumulant3 } \mu_N \text{ trU trU trU}$, and `kappa3Re` is the same with $\text{Re Tr } U$. Lean proves the moment formula $\kappa_3(N) = \mathbb{E}[X^3] - 3\mathbb{E}[X^2]\mathbb{E}[X] + 2\mathbb{E}[X]^3$ (`kappa3_eq_moments`). Lean also proves that every moment $\mathbb{E}[X^a \bar{X}^b]$ is a genuine integral (`integrable_trU_moment`: Tr is continuous on a compact group and μ_N is finite).

3 Proof

Lemma 5 (`integral_eq_zero_of_odd`). Let σ be a map with $\int f \circ \sigma d\mu = \int f d\mu$ for all f . If $f \circ \sigma = -f$, then $\int f d\mu = 0$.

Proof. $\int f = \int f \circ \sigma = \int (-f) = -\int f$. This uses only linearity of the Bochner integral under negation, which holds without integrability. $\square$

Lemma 6 (`cumulant3_eq_zero_of_odd`). If moreover $X \circ \sigma = -X$, $Y \circ \sigma = -Y$ and $Z \circ \sigma = -Z$, then $\kappa(X, Y, Z) = 0$.

Proof. XYZ , X , Y and Z are all odd under σ , so their integrals vanish. Every term of the joint cumulant formula contains one of these integrals as a factor. $\square$

Theorem 7 (`cumulant3_linear_trace_eq_zero_right`). Let μ be a left-invariant (resp. right-invariant) measure on $U(N)$, $N \geq 0$, and let $L_1, L_2, L_3 : \mathbb{C} \rightarrow \mathbb{K}$ be $\mathbb{R}$ -linear, with $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$. Then $\kappa(L_1 \text{Tr } U, L_2 \text{Tr } U, L_3 \text{Tr } U) = 0$.

Proof. The scalar matrix -1 lies in $U(N)$. Left invariance gives $\int f((-1)U) d\mu = \int f d\mu$ (`Mathlib.integral_mul_left_eq_self`). Right invariance gives the same with $U(-1)$, and $U(-1) = (-1)U$ because -1 is central. Next, $\text{Tr}((-1)U) = \text{Tr}(-U) = -\text{Tr } U$, so $L_k(\text{Tr}((-1)U)) = -L_k(\text{Tr } U)$. Apply the previous lemma with $\sigma(U) = (-1)U$. $\square$

Specializing $\mu = \mu_N$ and $L_k \in \{\text{id}, \bar{\cdot}, \text{Re}\}$ gives `kappa3_eq_zero`: $\kappa_3(N) = 0$ for all N . It also gives `mixed_cumulants_eq_zero` ($\kappa(X, X, \bar{X}) = \kappa(X, \bar{X}, \bar{X}) = 0$) and `kappa3Re_eq_zero`.

Theorem 8 (`kappa3_not_equivalent_conjecture_1870_false`). For every $c \in \mathbb{C} \setminus \{0\}$ and every $\alpha \in \mathbb{R}$, $\kappa_3(N) \not\sim c N^{-\alpha}$. In particular $\kappa_3(N) \not\sim c_3/N$, and $N\kappa_3(N) \not\rightarrow c_3$, for $c_3 = 2\pi i/3$.

Proof. κ_3 is the zero sequence. If $0 \sim v$, then $v \sim 0$, which forces $v_N = 0$ eventually (`Mathlib.isEquivalent_zero_iff_eventually_zero`). But $cN^{-\alpha} \neq 0$ for $N \geq 1$. For the limit: $N\kappa_3(N) = 0 \rightarrow 0 \neq c_3$, and limits in $\mathbb{C}$ are unique. $\square$

Here $u \sim v$ is `Mathlib's Asymptotics.IsEquivalent`, meaning $u - v = o(v)$. For $v = c/N$ with $c \neq 0$ this is exactly $\kappa_3(N) = c/N + o(1/N)$.

4 Lean formalization and axiom audit

File `lean/Conjecture1870/Basic.lean` (202 lines, Lean 4.33.1, Mathlib v4.33.1, only import Mathlib). Main theorem, transliterated to ASCII (the source writes `Not` as $\neg$, `/\` as $\wedge$, `Nat` and `Complex` as $\mathbb{N}$ and $\mathbb{C}$, and `nhds` as the neighbourhood-filter symbol):

```
theorem conjecture_1870_false :
  Not (kappa3 ~[atTop] fun N : Nat => c3 / N) /\
  Not (Tendsto (fun N : Nat => (N : Complex) * kappa3 N) atTop (nhds c3))

with c3 = 2 * Real.pi * Complex.I / 3. Further theorems: kappa3_not_equivalent,
cumulant3_linear_trace_eq_zero, cumulant3_linear_trace_eq_zero_right, kappa3_eq_
zero, kappa3_eq_moments, mixed_cumulants_eq_zero, kappa3Re_eq_zero, integrable_trU_
moment and isCompact_unitaryGroup. lake build succeeds. For every theorem listed above,
#print axioms reports [propext, Classical.choice, Quot.sound]. There is no sorry, no
native_decide, and no new axiom.
```

5 Relation to the earlier submission

PR #234 (orionsheep, closed) had the same, mathematically sound, rotation-symmetry argument. Its Lean was vacuous. In the reviewer’s words: “the capstone theorem is literally $(0:\text{Nat}) = 0 := \text{rfl}$ and the ‘zero moments’ are substituted numeric literals; no unitary matrix, Haar measure, trace, or cumulant appears in Lean, and the entire (mathematically sound) rotation-symmetry argument is prose-only.” This submission puts every one of those objects in Lean. It uses the group $U(N)$ as Mathlib’s `Matrix.unitaryGroup`, with its topology and Borel σ -algebra. It proves that $U(N)$ is compact, and builds the Haar probability measure `haarMeasure` $\top$. The trace is `Matrix.trace` and the cumulant is defined from Bochner integrals. The symmetry argument (invariance under $U \mapsto -U$) is a Lean proof for every N and for every left- or right-invariant measure. The asymptotic claim is refuted in Lean with `Asymptotics.IsEquivalent` and `Tendsto`.